KL UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Smart Campus Navigation, Facility Monitoring & Resource Analytics System (SCN-FMRA)

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CERTIFICATE

This is to certify that the project based laboratory report entitled "Smart Campus Navigation, Facility Monitoring & Resource Analytics System (SCN-FMRA)" submitted by **D Karthikeya** bearing Regd. No. 2500031413 to the **Department of BASIC ENGINEERING SCIENCES , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in "Python Programming (25SC2007)" course in I B Tech II Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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ABSTRACT

The Smart Campus Navigation, Facility Monitoring and Resource Analytics System (SCN-FMRA) is an integrated solution designed to enhance the efficiency, accessibility, and management of modern campus environments. This system combines intelligent navigation, real-time facility monitoring, and data-driven resource analytics into a single platform.

The navigation module helps users find the shortest and most efficient paths between different locations within the campus using graph-based algorithms. The facility monitoring component tracks real-time parameters such as occupancy levels, power status, and availability of campus resources, enabling better utilization and management of infrastructure. Additionally, the system collects and analyzes usage data to generate insights that assist administrators in decision-making and resource optimization.

By leveraging technologies such as Python programming, data structures, and basic simulation of IoT-based inputs, SCN-FMRA provides a scalable and cost-effective approach to smart campus management. This system improves user convenience, enhances operational efficiency, and supports the development of intelligent and sustainable campus ecosystems.

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INTRODUCTION

With the rapid growth of technology and increasing campus sizes, managing navigation, infrastructure, and resource utilization has become a challenging task for educational institutions. Students, staff, and visitors often face difficulties in locating buildings, accessing facilities, and understanding the availability of campus resources. At the same time, administrators require efficient systems to monitor and manage facilities effectively.

The Smart Campus Navigation, Facility Monitoring and Resource Analytics System (SCN-FMRA) is designed to address these challenges by integrating multiple functionalities into a unified platform. This system provides intelligent navigation to help users find optimal routes within the campus, ensuring ease of movement and time efficiency. It also includes a facility monitoring module that tracks important parameters such as occupancy levels and operational status of various campus facilities.

In addition, the system incorporates resource analytics to collect and analyze usage data, enabling better decision-making and efficient allocation of resources. By using technologies like Python programming, graph algorithms, and simulated sensor data, the project demonstrates how smart solutions can improve campus management.

Overall, SCN-FMRA aims to create a smarter, more efficient, and user-friendly campus environment by combining navigation, monitoring, and analytics into a single system.

AIM

The aim of the **Smart Campus Navigation, Facility Monitoring and Resource Analytics System (SCN-FMRA)** is to design and develop an intelligent and integrated platform that improves the overall management and usability of a campus environment. The system focuses on providing efficient navigation services to help students, staff, and visitors easily locate different buildings and facilities within the campus using optimized path-finding techniques.

Another important aim of this project is to monitor campus facilities in real time by tracking parameters such as occupancy levels, availability, and operational status. This helps in better utilization of infrastructure and ensures that resources are used effectively without overcrowding or wastage.

Furthermore, the project aims to implement resource analytics by collecting and analyzing usage data of various facilities. These insights assist administrators in making informed decisions regarding maintenance, resource allocation, and future planning.

Overall, SCN-FMRA aims to create a smart, efficient, and user-friendly campus system that enhances convenience, improves resource management, and supports the development of a modern digital campus.

ADVANTAGES

- **Easy Navigation**
- **Real-Time Facility Monitoring**
- **Efficient Resource Utilization**
- **Data-Driven Decision Making**
- **Improved User Experience**
- **Cost-Effective Solution**

DISADVANTAGES

- **High Initial Setup Cost**
- **Dependence on Technology and Internet**
- **Maintenance and Upgradation Required**
- **Data Privacy and Security Concerns**
- **System Complexity**
- **Possible Sensor/Data Inaccuracy**

FUTURE IMPLEMENTATION

- Use **IoT sensors** for real-time data collection
- Develop a **mobile app** for easy access
- Add **cloud storage** for better data management
- Implement **AI/ML** for smart predictions

SYSTEM REQUIREMENTS

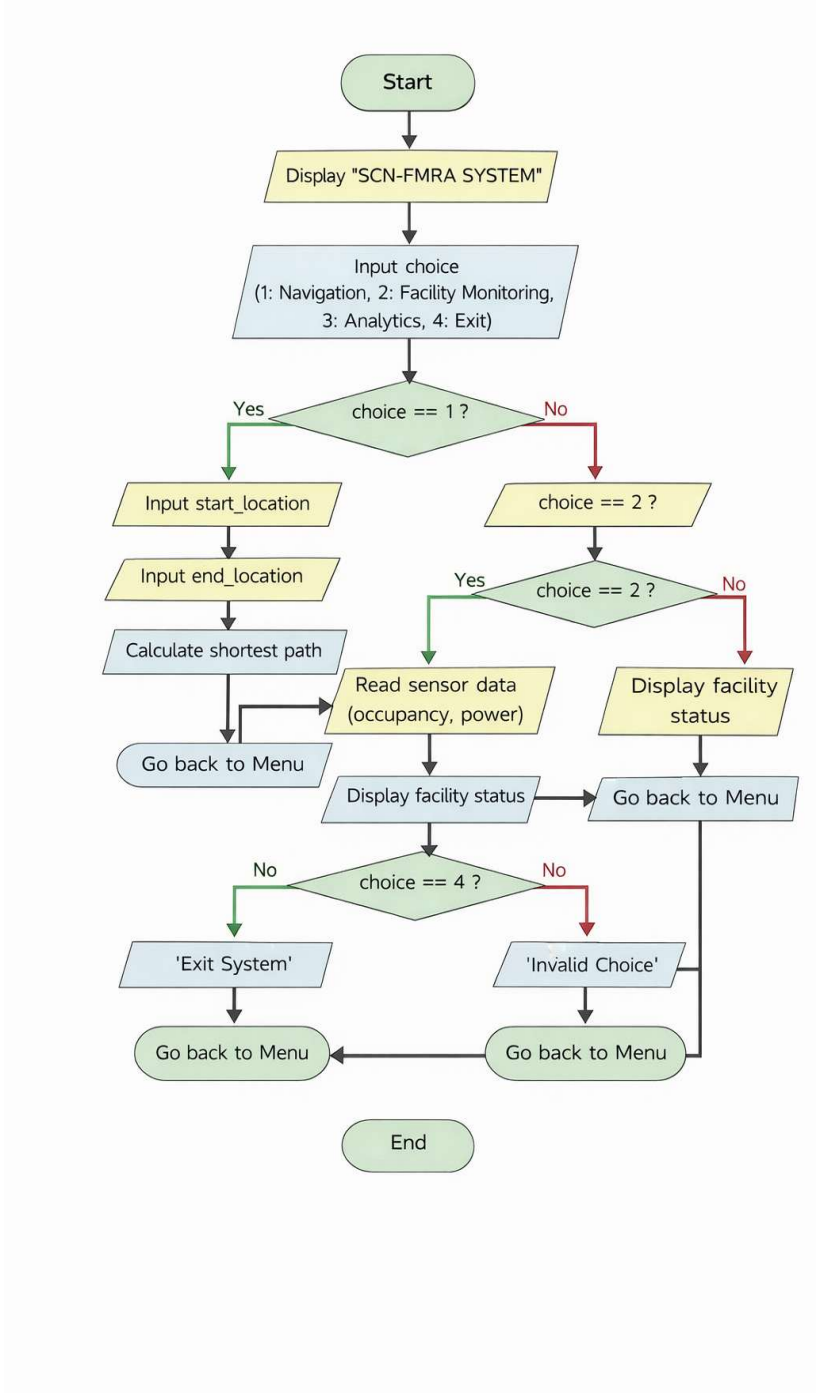
➢ SOFTWARE REQUIREMENTS:

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **IDE/Editor:** PyCharm / VS Code / IDLE
- **Database:** SQLite / MySQL
- **Libraries:**
  - ➢ heapq (for path finding)
  - ➢ tkinter (for GUI)
  - ➢ matplotlib (for analytics/graphs)
- **Web Framework (optional):** Flask / Django
- **Browser (if web-based):** Chrome / Edge / Firefox

➢ HARDWARE REQUIREMENTS:

- **Computer/Laptop** (for development and running the system)
- **Microcontroller** (Arduino / Raspberry Pi – for IoT integration)
- **Sensors** (e.g., occupancy sensor, temperature sensor)
- **Wi-Fi Module / Internet Connection**
- **Power Supply Unit**
  - ➢ • **Display Device** (monitor or mobile device for user interface)

DATA FLOW DIAGRAM



ALGORITHM

Step 1: Start

Step 2: Display Menu

- Show options:
    1. Navigation
    2. Facility Monitoring
    3. Resource Analytics
    4. Exit

Step 3: Read User Choice

Step 4: Check User Choice

Case 1: Navigation

1. Input start location
2. Input destination
3. Apply shortest path algorithm (Dijkstra)
4. Display shortest path and distance
5. Go back to menu

Case 2: Facility Monitoring

1. Read sensor data (occupancy, power status)
2. Update facility information
3. Display facility status
4. Go back to menu

Case 3: Resource Analytics

1. Collect usage data
2. Analyze data (count usage frequency)
3. Generate report
4. Display analytics results
5. Go back to menu

Case 4: Exit

1. Display exit message
2. Stop execution

Invalid Choice

1. Display error message
2. Return to menu

Step 5: Repeat Until Exit

Step 6: End

IMPLEMENTATION

```
import heapq
import random
from datetime import datetime

# -------------------------------
# CAMPUS GRAPH (Navigation)
# -------------------------------
campus_map = {
"Gate": {"Admin": 2, "Library": 5},
"Admin": {"Gate": 2, "Canteen": 4, "Lab": 6},
"Library": {"Gate": 5, "Lab": 3},
"Canteen": {"Admin": 4, "Hostel": 2},
"Lab": {"Admin": 6, "Library": 3, "Hostel": 4},
"Hostel": {"Canteen": 2, "Lab": 4}
}

# -------------------------------
# DIJKSTRA (Shortest Path)
# -------------------------------
def shortest_path(graph, start, end):
pq = [(0, start, [])]
visited = set()

while pq:
(cost, node, path) = heapq.heappop(pq)

if node in visited:
continue

path = path + [node]
visited.add(node)

if node == end:
return cost, path

for neighbor, weight in graph[node].items():
```

```
heapq.heappush(pq, (cost + weight, neighbor, path))

return float("inf"), []

# -------------------------------
# FACILITY MONITORING
# -------------------------------
facilities = {
"Library": {"occupancy": 0, "power": "ON"},
"Lab": {"occupancy": 0, "power": "ON"},
"Canteen": {"occupancy": 0, "power": "ON"},
"Hostel": {"occupancy": 0, "power": "ON"},
}

def update_facilities():
for place in facilities:
facilities[place]["occupancy"] = random.randint(0, 100)
facilities[place]["power"] = random.choice(["ON", "OFF"])

def display_facilities():
print("\n--- Facility Status ---")
for place, data in facilities.items():
print(f"{place}: Occupancy={data['occupancy']}%, Power={data['power']}")

# -------------------------------
# RESOURCE ANALYTICS
# -------------------------------
usage_log = []

def log_usage(location):
usage_log.append((location, datetime.now()))

def show_analytics():
print("\n--- Resource Analytics ---")
count = {}
for loc, _ in usage_log:
count[loc] = count.get(loc, 0) + 1

for loc, c in count.items():
```

```
print(f"{loc} used {c} times")

# -------------------------------
# MAIN SYSTEM MENU
# ------------------------------
def main():
while True:
print("\n=== SCN-FMRA SYSTEM ===")
print("1. Find Shortest Path")
print("2. Show Facility Status")
print("3. Log Facility Usage")
print("4. Show Analytics")
print("5. Exit")

choice = input("Enter choice: ")

if choice == "1":
start = input("Enter start location: ")
end = input("Enter destination: ")

if start in campus_map and end in campus_map:
cost, path = shortest_path(campus_map, start, end)
print(f"Shortest Path: {' -> '.join(path)}")
print(f"Distance: {cost}")
else:
print("Invalid location!")

elif choice == "2":
update_facilities()
display_facilities()

elif choice == "3":
loc = input("Enter facility name: ")
if loc in facilities:
log_usage(loc)
print("Usage logged!")
else:
print("Invalid facility!")
```

```
        elif choice == "4":
        show_analytics()

        elif choice == "5":
        print("Exiting system...")
        break

        else:
        print("Invalid choice!")

        # -------------------------------
        # RUN PROGRAM
        # -------------------------------
        if __name__ == "__main__":
main()
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
=== SCN-FMRA SYSTEM ===
 1. Find Shortest Path
 2. Show Facility Status
 3. Log Facility Usage
 4. Show Analytics
 5. Exit
Enter choice: 1
Enter start location: Gate
Enter destination: Hostel
Shortest Path:  Gate -> Admin -> Canteen -> Hostel
Distance: 8
```

```
=== SCN-FMRA SYSTEM ===
 1. Find Shortest Path
 2. Show Facility Status
 3. Log Facility Usage
 4. Show Analytics
 5. Exit
Enter choice: 2

 --- Facility Status ---
 Library:   Occupancy= 73%, Power=ON
 Lab:       Occupancy= 25%, Power=OFF
 Canteen:   Occupancy= 91%, Power=ON
 Hostel:    Occupancy= 40%, Power=OFF
```

```
=== SCN-FMRA SYSTEM ===
 1. Find Shortest Path
 2. Show Facility Status
 3. Log Facility Usage
 4. Show Analytics
 5. Exit
Enter choice: 3
Enter facility name: Library
Usage logged!

Enter choice: 3
Enter facility name: Lab
Usage logged!

Enter choice: 4
```

```
--- Resource Analytics ---
Library used 2 times
Lab used 1 times
Canteen used 1 times
Hostel used 0 times

=== SCN-FMRA SYSTEM ===
 1. Find Shortest Path
 2. Show Facility Status
 3. Log Facility Usage
 4. Show Analytics
 5. Exit
Enter choice: 5
Exiting system...
```

CONCLUSION

The **Smart Campus Navigation, Facility Monitoring and Resource Analytics System (SCN-FMRA)** successfully integrates path-finding algorithms with real-time data tracking to modernize campus management. By automating navigation and facility oversight, the system reduces manual effort and improves operational efficiency.